
Quantifying “Chill” Affect: The Case of Bisexual Stoners

Bayleigh Baldwin (she/they)
Linguistics PhD Student, The University of Kansas

Why bisexual stoners?

We know little about how covariation functions with multiple sociophonetic variables in general

- This is a great test case for seeing how the same linguistic variables can index a “chill” stoner persona and bisexuality simultaneously

Research question: How do bisexual stoner women perform their gender identity (womanhood), gender expression (e.g. femininity), and sexual orientation (bisexuality) through language?

Literature review: creak

Studies have looked at creak as indexical of sexual identity and gender expression but results are mixed for queer women:

- One production study found no difference between straight women, bisexual women, and lesbian women's use of creak (Barron-Lutzross, 2018)
- Another found that femme lesbians use less creak than masc lesbians (Sulkin & Tighe, 2025)
- More creak can be perceived as generally lesbian (Barron-Lutzross, 2018)
- Less creak can be perceived as more queer for women (Herubin, 2025)

Literature review: creak and “chill” affect

In Pratt (2023), high school boys in California who were self-described as “chill” or “stoners” used more creaky voice in comparison to the “high energy” boys



Literature review: pitch (related to creak?)

In gay topics, lower pitch used by butches, higher pitch by femmes

What are bisexual women doing?

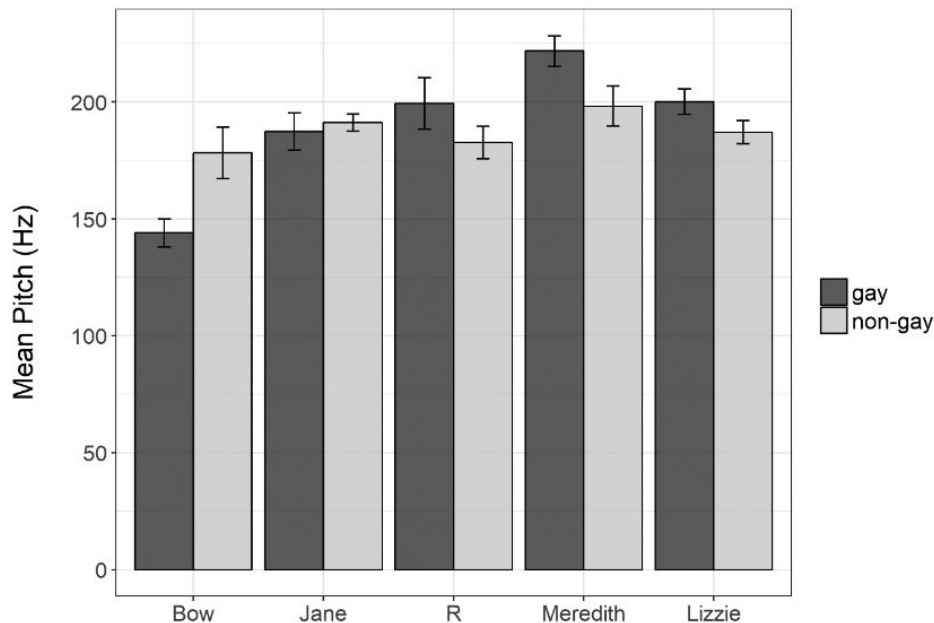
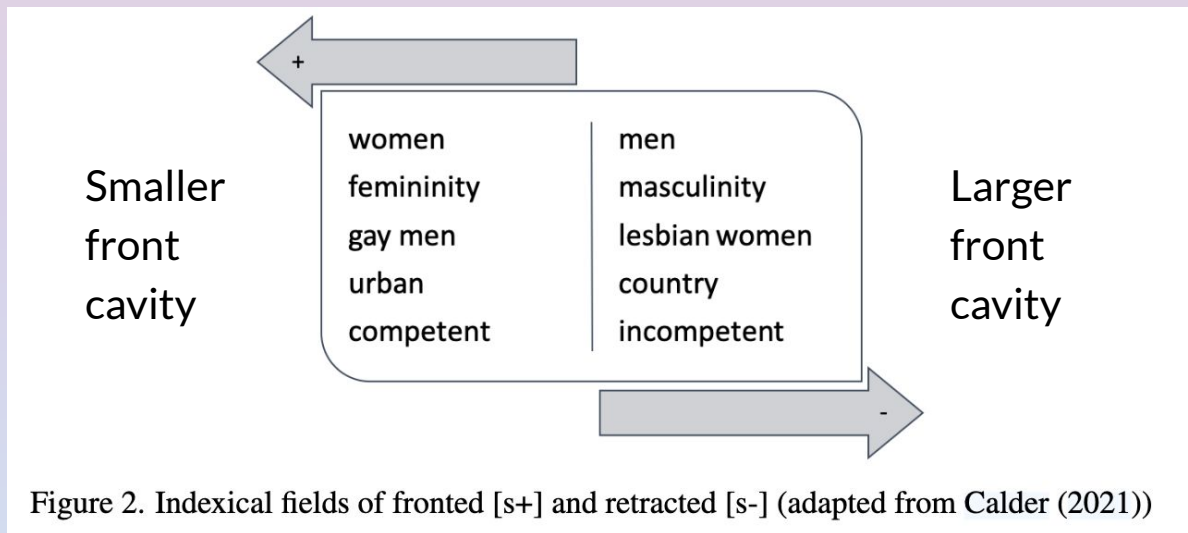


Figure 3.2 Average mean pitch level in Hz for speakers ordered from left to right by position on the 'butch'-'lipstick' continuum and divided by speech topic.

Literature review: /s/

Sibilant sounds are a group of linguistic variables with salient social meanings



Where do bisexual women fit?

Literature review: /s/

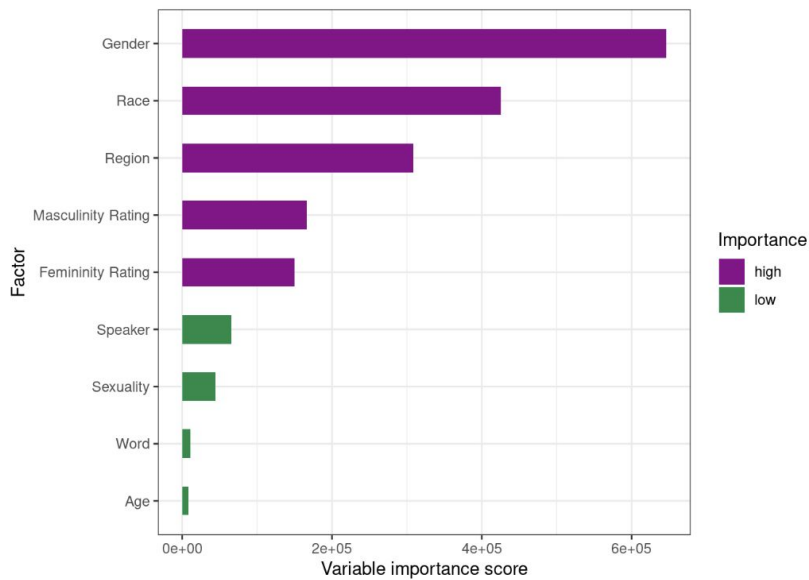


Figure 3. COG VIMP scores

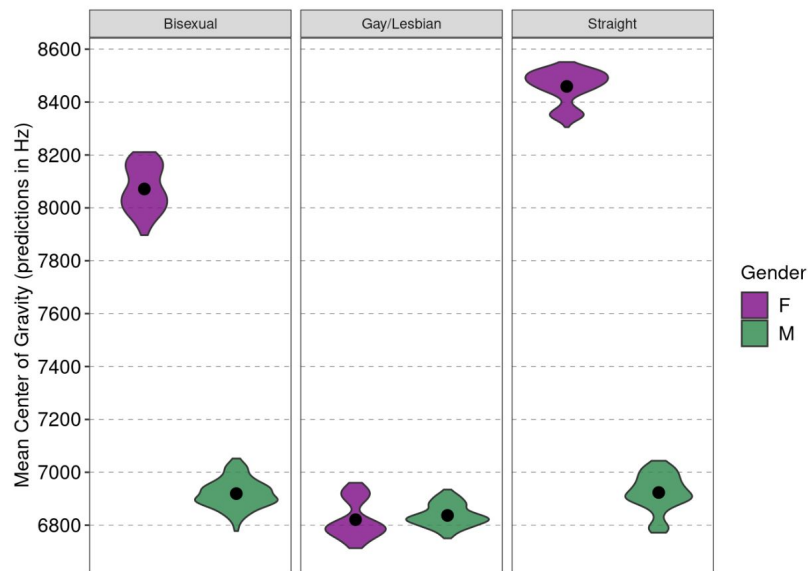


Figure 7. GENDER:SEXUALITY partial dependence co-plot

Sexuality perhaps isn't the most important factor influencing /s/ production!

Predictions

1. Bisexual stoner women will use a lot of creaky voice, and will use most creak when talking about weed and queerness (non-normative femininity)
2. Bisexual stoner women will have retracted /s/ when talking about weed (masculine?) and a fronter /s/ when talking about queer topics (feminine sexual identity)

The current case study

- Sociolinguistic interviews of two bisexual women who identify with femininity, aged 22 and 23, from Kansas City and the Bay Area of California
- Central topics elicited: childhood, queerness, weed (generally in this order)
- Also collapsed into Gay and Non-gay topics like Lawrence (2014) (cited within Levon, 2021) but this did not give enough nuance

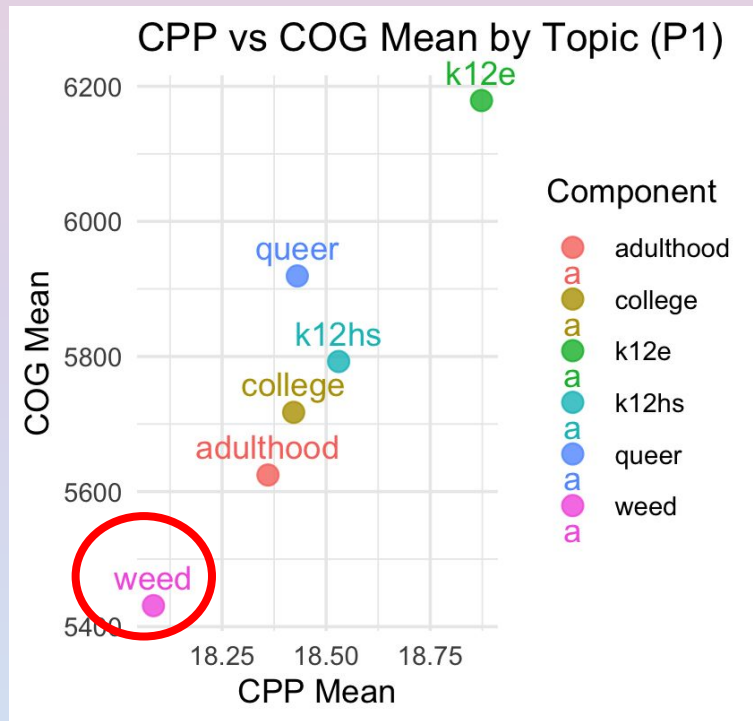
Methods

- Creak measurements:
 - Used PraatSauce (Kirby & Puggaard-Rode, 2025) to get measures of spectral tilt (H1-H2 and CPP) and f0 at 5 equidistant points for every vowel, then averaged all 5 points to have one value
 - CPP used in analysis because it doesn't include breathy voice like H1-H2 and patterned with f0
- Sibilance measurements:
 - Noise reduction of background noise performed
 - Spectral moments of /s/ measured in Praat with 6 windows, 15 ms each then averaged to get COG and skew, also logged duration (DiCanio 2013)

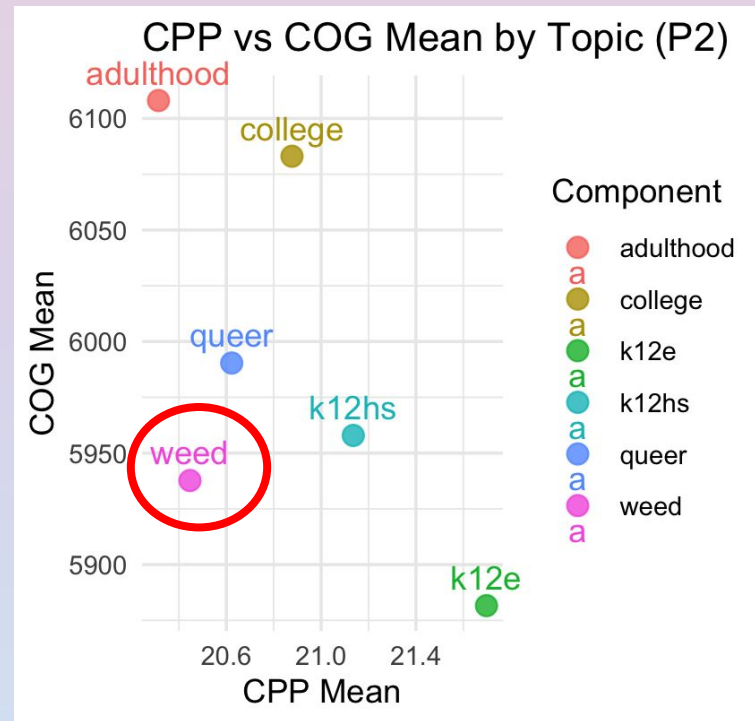
Scatter plot of topics

Frontier /s/

Backer /s/



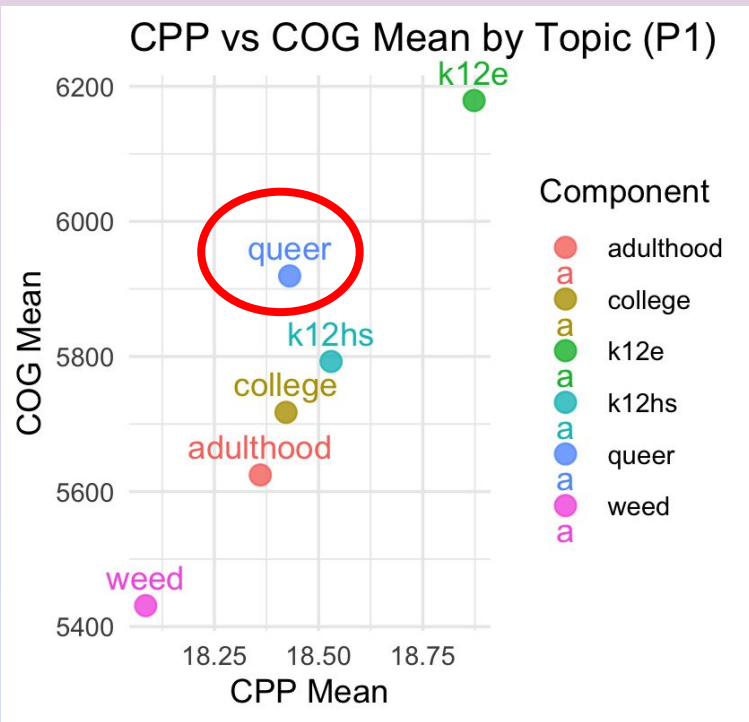
More creak Less creak



More creak Less creak

Scatter plot of topics

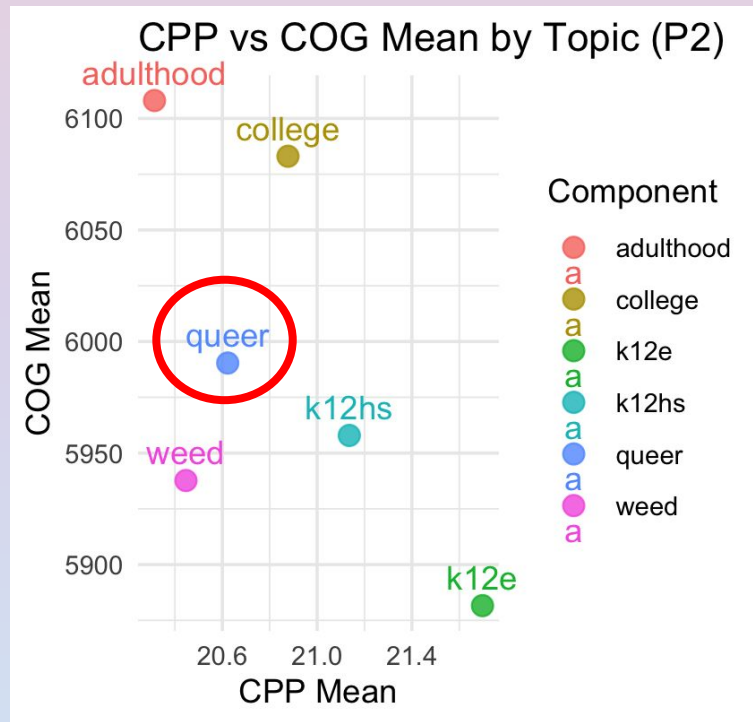
Frönter /s/



Backer /s/

More creak

Less creak

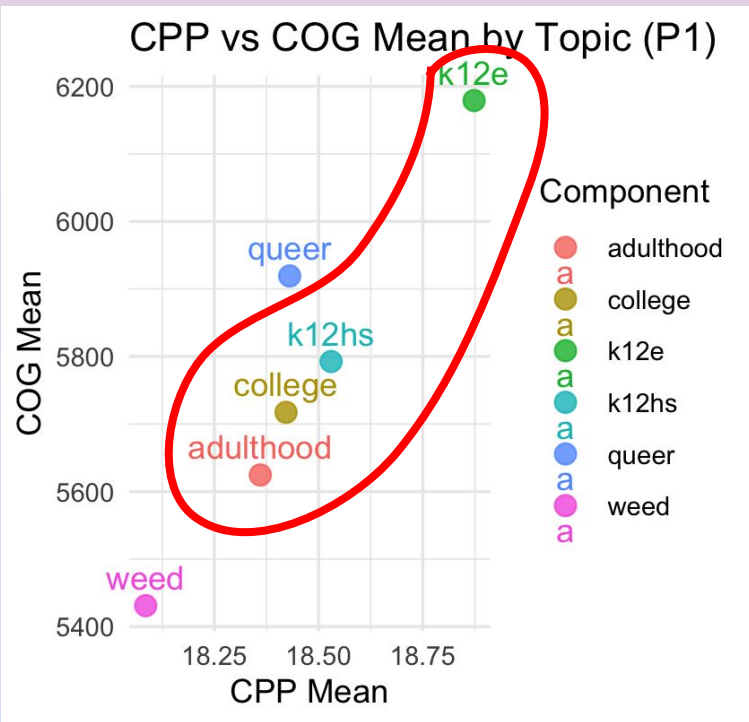


More creak

Less creak

Scatter plot of topics

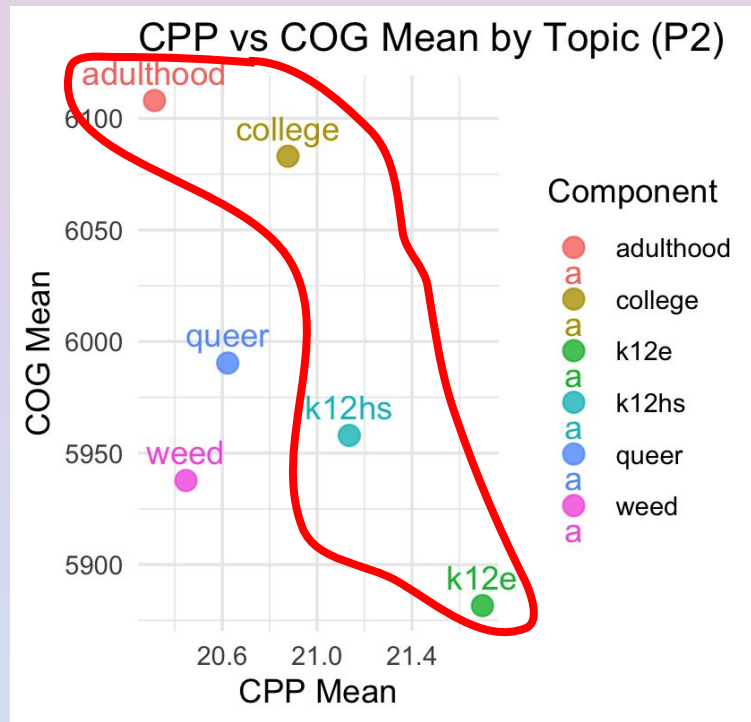
Frontier /s/



More creak

Less creak

Backer /s/



More creak

Less creak

Future Plans

- Will hand code creak
- Plan to get 10 participants total and compare with other groups (male stoners? non bisexuals?)
- Queer women /s/ length as a part of cluster of features? (Lang, 2023)
- What about stoned speech?
- What about effect of time course of the interview?
- Style shifting in different contexts, like work vs home

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